Session 7.
Standardization and the Normal Distribution.

Pre course: Statistics.

Pablo Herrera G.

Hochschule
Bonn-Rhein-Sieg
University of Applied Sciences

Agenda

- Z-scores.
- Density curves.
- Normal distribution.
- Applied exercise.
- Empirical rule.
- Normal distribution calculations.

Agenda

- **Z-scores.**
- Density curves.
- Normal distribution.
- Applied exercise.
- Empirical rule.
- Normal distribution calculations.

Z-scores.

General idea:

- Give us the number of std deviations from the population mean for a particular point.
- We can analyze how many standard deviations an observation is above or below the mean.
- Tell us how usual or unusual an observation is.

$$Z = \frac{X - \mu}{\sigma}$$

How can this be useful?

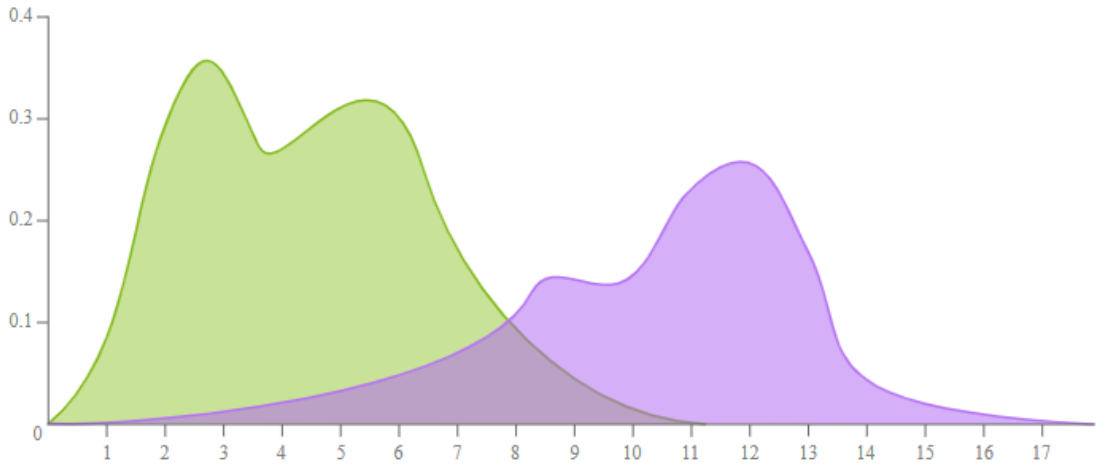
- When two variables have different scales, we can not just look at the absolute values.
- We can look at the Z-scores instead.

Agenda

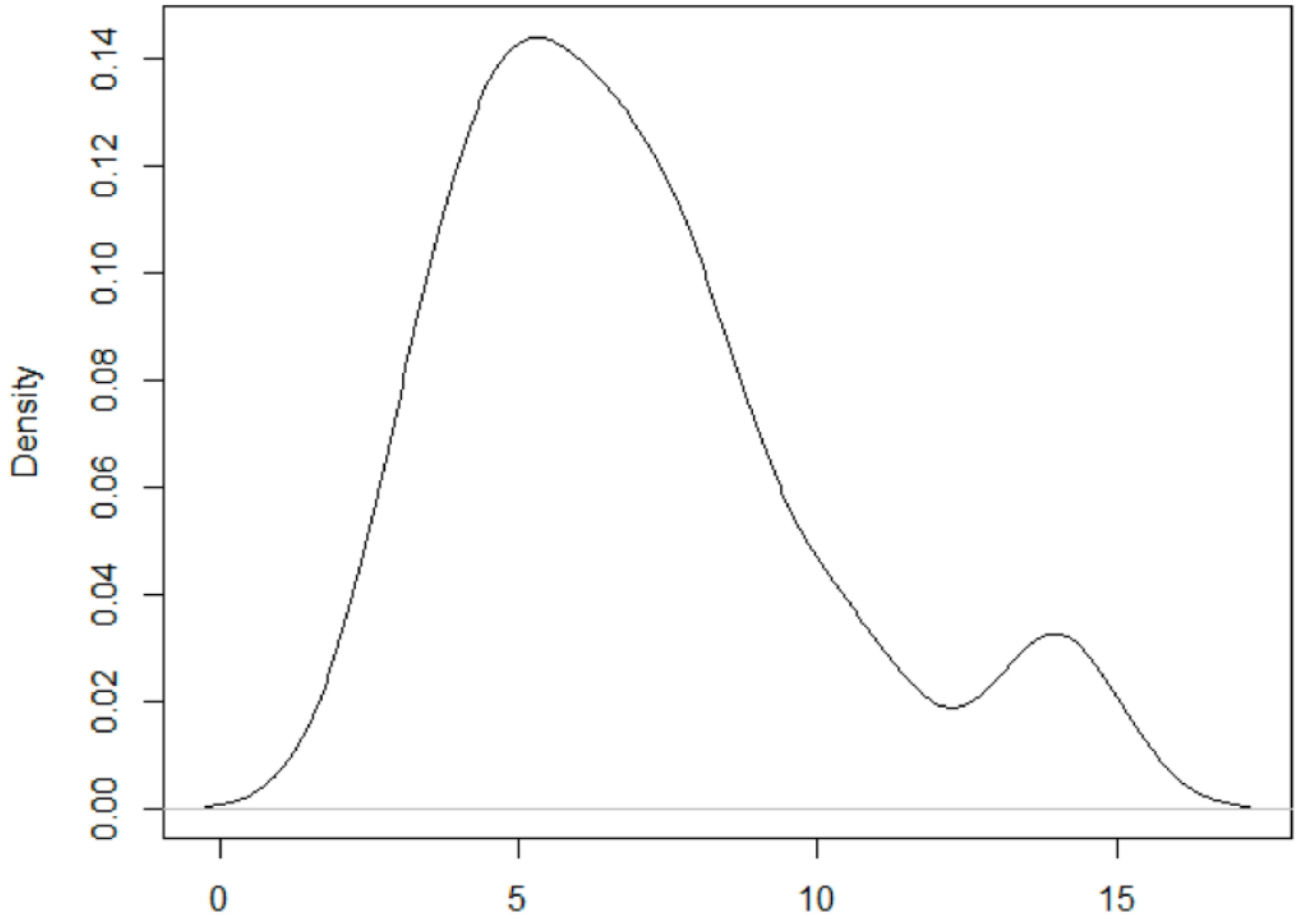
- Z-scores.
- **Density curves.**
- Normal distribution.
- Applied exercise.
- Empirical rule.
- Normal distribution calculations.

Density curves.

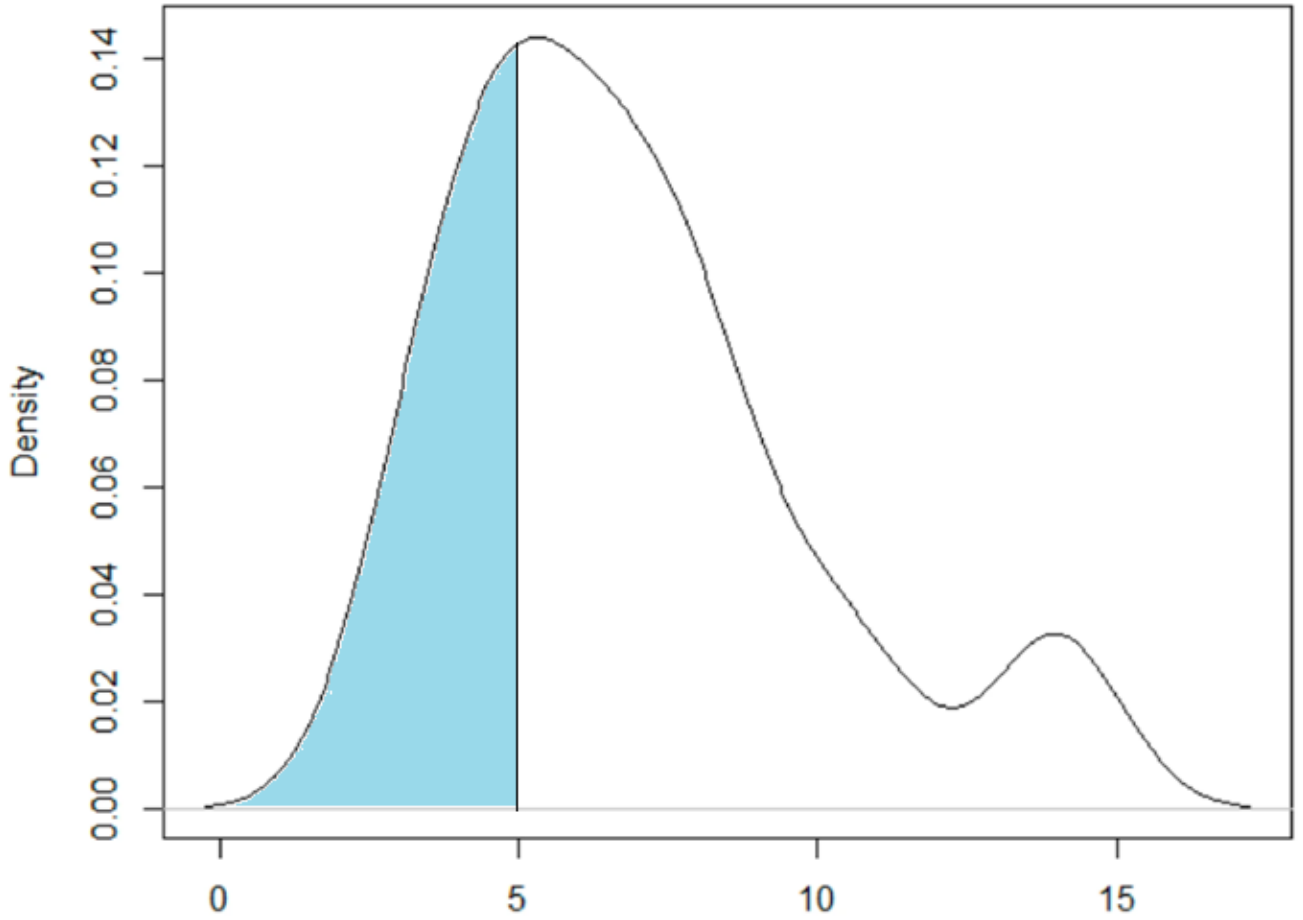
- They are smoothed graphical representations of the distribution of a continuous variable.
- We talk about percentage of data within a range (area under the curve within that interval).



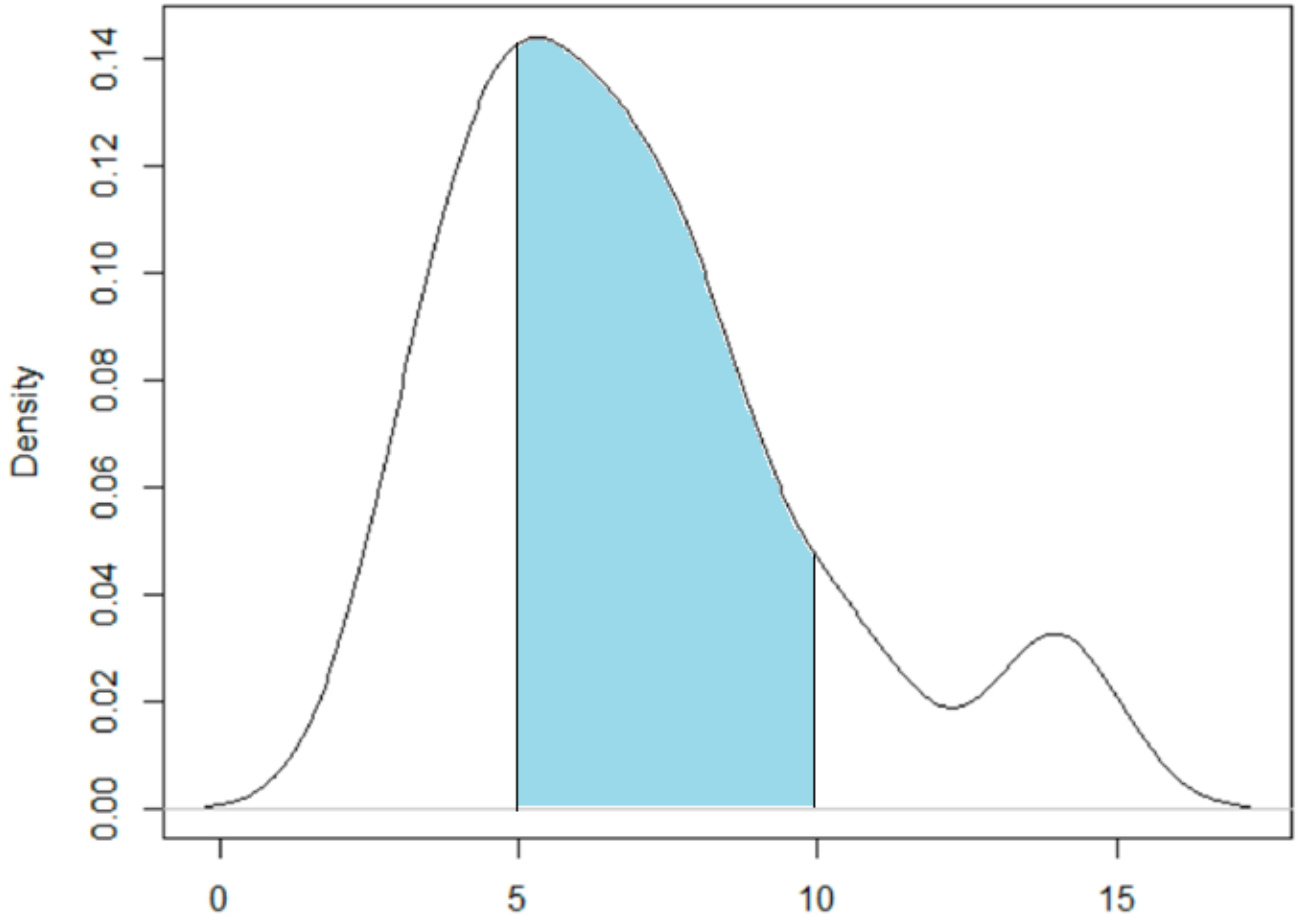
Density curves.



Density curves.



Density curves.

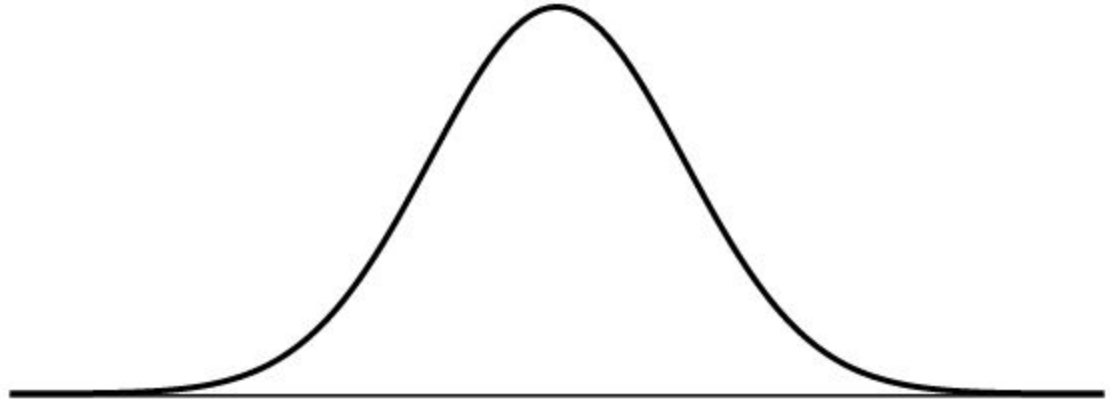


Agenda

- Z-scores.
- Density curves.
- **Normal distribution.**
- Applied exercise.
- Empirical rule.
- Normal distribution calculations.

Normal distribution.

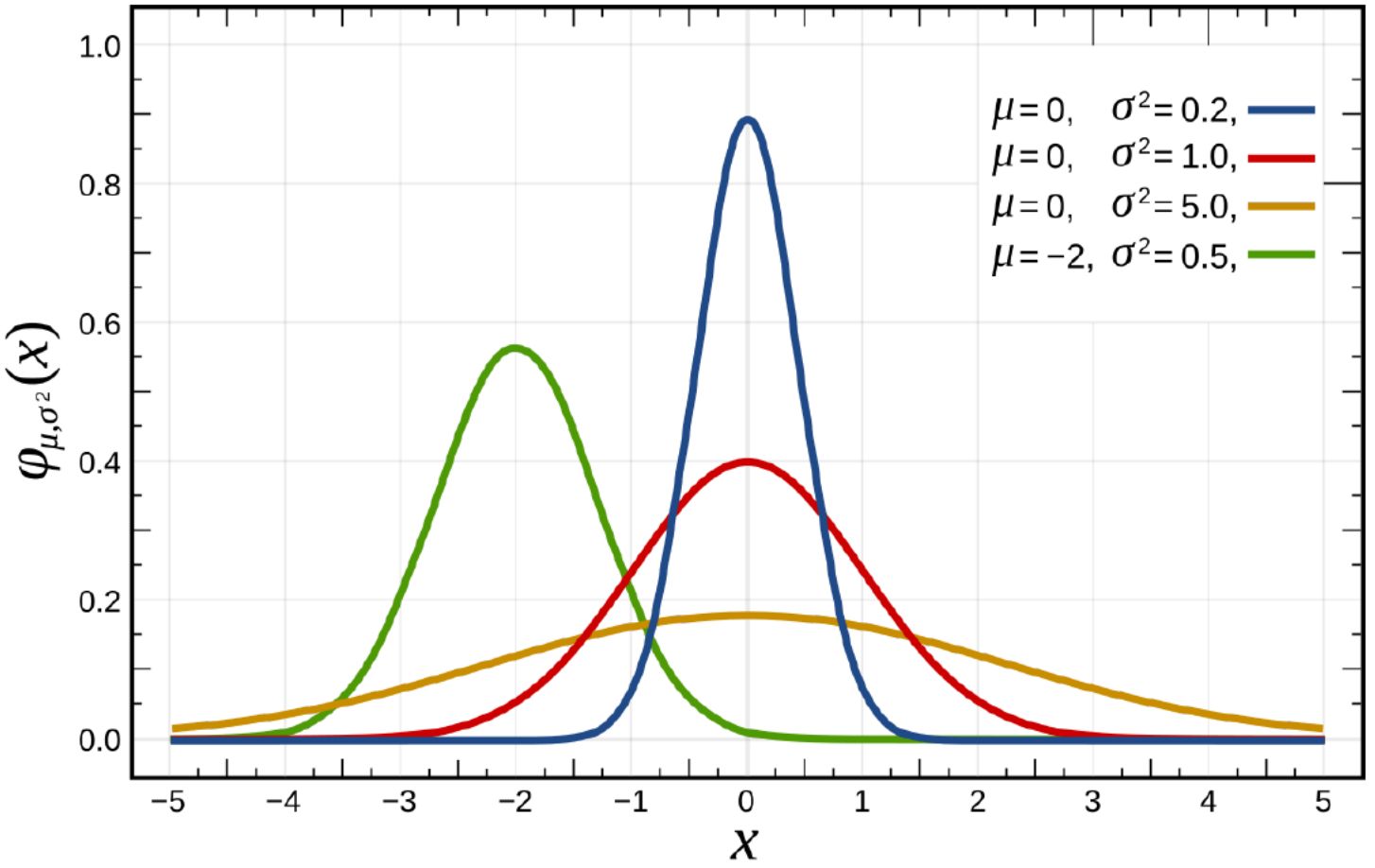
- It is a probability distribution that is symmetric around its mean.
- Same mean, median, and mode. At the center of the distribution.
- Bell shaped, unimodal, and symmetric.



Normal distribution.

Controlled by two parameters:

- μ = Mean.
- σ = Standard deviation.



Normal distribution.

- Examples of variables that **could** distribute like a normal.
  - Health quality indicators.
  - Educational test scores.
  - Work hours within some populations.
  - Age of beneficiaries of a program.
  - Energetic consumption.
  - Height and weight of new borns.
- Examples of variables that **might not** distribute like a normal
  - Income.
  - Unemployment period.
  - Health care spending.

Agenda

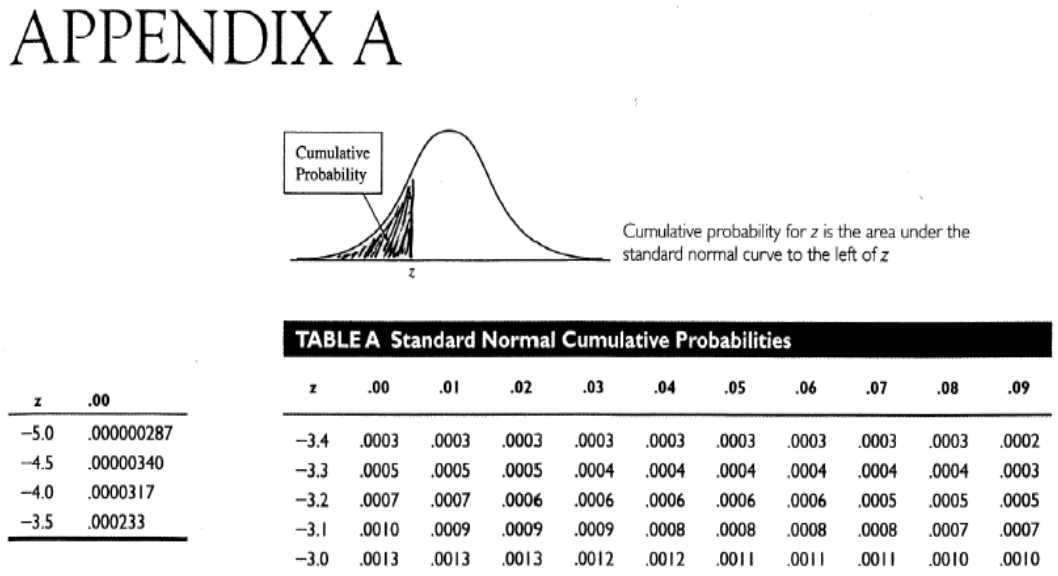
- Z-scores.
- Density curves.
- Normal distribution.
- **Applied exercise.**
- Empirical rule.
- Normal distribution calculations.

Applied exercise.

The Table of Standard Normal Cumulative Probabilities provides the cumulative probability of various z-scores for a standard normal distribution ($\mu = 0$, $\sigma = 1$), also called Normal(0,1).

Calculate the probability of being under:

- 3 Std. deviations.
- 2 Std. deviations.
- 1 Std. deviations.



APPENDIX A

Cumulative probability for z is the area under the standard normal curve to the left of z

z	.00
−5.0	.000000287
−4.5	.00000340
−4.0	.0000317
−3.5	.000233

TABLE A Standard Normal Cumulative Probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010

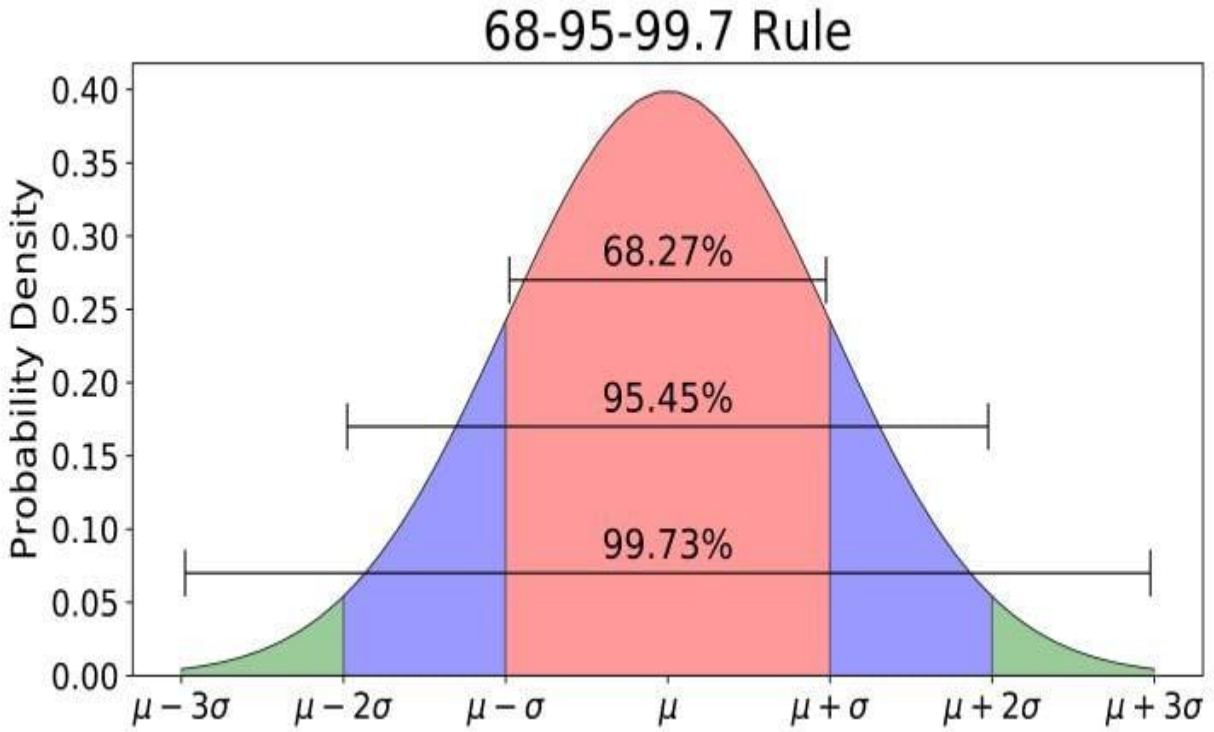
Agenda

- Z-scores.
- Density curves.
- Normal distribution.
- Applied exercise.
- **Empirical rule.**
- Normal distribution calculations.

Empirical rule.

The 68 – 95 – 99,7 rule.

- Quick way to estimate probabilities for a normal distribution.
- Shows how concentration of data happens around the mean.
- Useful for checking whether an observation is “typical” or “unusual.”



Agenda

- Z-scores.
- Density curves.
- Normal distribution.
- Applied exercise.
- Empirical rule.
- **Normal distribution calculations.**

Normal distribution calculations.

- Standard normal table for proportion under or above.
- Standard normal table for proportion between values.
- Finding z-score for a percentile (opposite way).